L8b: Option Contracts: Call and Put Payoff and Profit

CHEME 5660 Cornell University

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today opens Unit 3. Units 1 and 2 modeled assets we hold directly; a derivative is instead a contract whose cash flows are a function of someone else's price. We read the contract, compute what it pays at expiration, and keep what the contract promises separate from what the investor earns.

Today's concepts

- Read an option contract: underlying, strike K, expiration T, exercise style, deliverable q, direction θ, and the premium: which are contractual, which is quoted.
- Separate payoff from profit: the expiration payoff of all four positions, the premium adjustment and the convention it hides, the breakeven prices, and the maximum profit and loss of each.
- Compose positions: a strategy as a signed sum of legs, with a corner at each strike where the legs do not cancel and an offset set by the net debit or credit.

Lectures and Examples

Lecture:
CHEME-5660-L8b-Lecture-IntroductionToDerivativesContracts-Fall-2026.ipynb

Example 1:
CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

The first example builds all four single-contract positions from a real options chain, sweeps the full domain $S_T \geq 0$, converts per-share quantities to per-contract dollars, and verifies every breakeven and bound numerically; the second combines signed legs into vertical spreads, straddles, and strangles.

Concept Review: From Assets to Contracts Written on Assets

Units 1 and 2 priced and allocated assets we hold directly: Treasury cash flows (L2a, L2b), equity growth rates (L3a, L4b), and portfolios built from them (L5b through L7b).

Carried forward: the price of the underlying. Let $S_t \geq 0$ be that price in USD/share at time t, so S_0 is today and S_T is expiration.

- The payoff is contractual, not statistical: given S_T, what the contract pays is a formula written into it. No estimation enters that step. Uncertainty enters only through S_T.
- The two sides are not symmetric: one counterparty holds a right, the other carries an obligation. The premium is the price of that asymmetry, and it is paid whether or not the right is ever used.

What Is a Derivative?

A derivative is a contract whose cash flows depend on an underlying quantity: an equity price, a rate, a currency, a commodity, an index, or a credit event. The families differ in who is obligated:

- Futures and forwards: *both* sides must transact the underlying later at a set price. Futures are exchange-traded and standardized; forwards are negotiated over the counter.
- Swaps: *both* sides must exchange two streams of cash flows. Traded over the counter.
- Options: the buyer holds a *right*, and only the seller carries an obligation. Listed equity options are exchange-traded and centrally cleared; an over-the-counter market also exists.

That asymmetry is the subject of the lecture: it makes an option payoff a kinked function instead of a straight line, and it is what the premium buys.

Reading an Option Contract

Seven quantities must be pinned down before any payoff can be computed:

- the underlying asset
- $K > 0$: strike, USD/share
- $T > 0$: expiration, years
- exercise style: European or American
- $q \in \mathbb{N}$: deliverable, shares/contract
- $\theta \in \{-1, +1\}$: direction, $+1$ long, -1 short
- $C_0, P_0 \geq 0$: call and put premium, USD/share

Only the premium is a market quantity: it is quoted and it moves, and L9a is where it comes from. The rest are contractual or chosen when you trade.

- Exercise style: European exercises only at T, American on or before T. At expiration the two behave identically, so it does not enter today; it affects the premium (L9a, L9b).
- The deliverable q: standard contracts use $q = 100$, but the specification controls. A premium quoted at 21.58 USD/share is $2{,}158$ USD of cash for one contract.

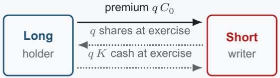
Reading an Option Contract: The Cash Flows

An option contract: a right and an obligation

The premium is paid once, at the open, whether or not the right is ever used

Call contract
THE RIGHT TO BUY AT K

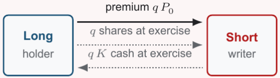
- **Long (holder)** pays C_0/share for the right to buy at K.
- **Short (writer)** collects C_0, must deliver at K if assigned.
- At T, exercised when $S_T > K$: payoff $(S_T - K)^+$.



solid: always. dotted: only if $S_T > K$.

Put contract
THE RIGHT TO SELL AT K

- **Long (holder)** pays P_0/share for the right to sell at K.
- **Short (writer)** collects P_0, must buy at K if assigned.
- At T, exercised when $S_T < K$: payoff $(K - S_T)^+$.



solid: always. dotted: only if $S_T < K$.

European style exercises only at T; American style on or before T. Commonly $q = 100$ shares/contract.

One flow is unconditional, the premium at the open; two are conditional on exercise. Long positions risk only the premium, short ones collect it and carry the obligation.

Expiration Payoff Is Contractual

For any real x write $x^+ \equiv \max(x, 0)$. With $S_T \geq 0$ the underlying price at expiration and $K > 0$ the strike, both in USD/share, the payoff per share to the long holder is given by:

$$\boxed{h_c(S_T; K) = (S_T - K)^+, \qquad h_p(S_T; K) = (K - S_T)^+}$$

The call holder exercises only when $S_T > K$, the put holder only when $S_T < K$, and otherwise abandons the right. Because the holder may always decline, both payoffs are nonnegative: the signature of a right.

- The short side mirrors it: with direction $\theta \in \{-1, +1\}$, the signed payoff is $\theta\, h_c$ or $\theta\, h_p$.
- Moneyness classifies exercise value, not profit: a call is in the money when $S_T > K$, out of the money when $S_T < K$, and the put reverses these. It says nothing about whether the trade made money.

Profit Requires a Cash-Flow Convention

Payoff belongs to the contract. Profit belongs to the investor, and it needs a rule for the premium, which moved at a different time. Under the undiscounted convention, with $C_0, P_0 \geq 0$ the premiums in USD/share and θ the direction, profit per share is given by:

$$\boxed{\Pi_T^{c,\theta}(S_T) = \theta\Big(\underbrace{h_c(S_T;K)}_{\text{payoff at } T} - \underbrace{C_0}_{\text{premium at } 0}\Big), \qquad \Pi_T^{p,\theta}(S_T) = \theta\Big(h_p(S_T;K) - P_0\Big)}$$

For a long position the premium is an outflow, for a short an inflow: one θ multiplies both terms.

- Say what it hides: this mixes time-0 and time-T dollars, which every earlier lecture avoided. A time-T P&L would grow the premium forward, a time-0 value would discount the payoff (L1b).
- Why use it: it gets the shape and the breakeven right with the fewest moving parts. A convention, not an accounting identity.

Profit: Where a Position Breaks Even

The breakeven S_{BE}, in USD/share, is the terminal price at which profit is exactly zero. Setting $\Pi_T = 0$ on the exercised branch gives $S_T - K - C_0 = 0$ for the long call and $K - S_T - P_0 = 0$ for the long put, hence:

$$\boxed{S^c_{\text{BE}} = K + C_0, \qquad S^p_{\text{BE}} = K - P_0}$$

- The breakeven is not the strike: the premium shifts the curve down (long) or up (short) without moving the kink at K. On the AMD chain a $K = 230$ call at $C_0 = 21.58$ breaks even at 251.58: finishing in the money is not the same as making money.
- Both directions share it: short profit is the exact negative of long profit, so both sides agree on the breakeven price.

Example 1: CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Loss Bounds and Direction Matter

All per share; multiply by q and by the number of contracts for account dollars.

Position	Maximum profit	Maximum loss	Breakeven
Long call	unbounded as $S_T \to \infty$	C_0	$K + C_0$
Short call	C_0	unbounded as $S_T \to \infty$	$K + C_0$
Long put	$K - P_0$ at $S_T = 0$	P_0	$K - P_0$
Short put	P_0	$K - P_0$ at $S_T = 0$	$K - P_0$

The difference between the call rows and the put rows is the domain: $S_T \geq 0$, so $h_p \leq K$ is capped by the strike, while h_c has no cap because S_T has none.

- Only one loss is genuinely unlimited: the uncovered short call. The long call's profit is unbounded too, but no loss is.
- A short put's loss is finite: $K - P_0$ at $S_T = 0$. On the $K = 220$ contract collecting 16.33, that is 203.67 USD/share, or $20{,}367$ USD per contract. Large is not unlimited.

Composite Positions Add Leg by Leg

Let position $\mathcal{C}$ contain legs $i = 1, \dots, d$ on one underlying and one expiration. Leg i has strike $K_i > 0$, direction $\theta_i \in \{-1, +1\}$, copies $n_i \in \mathbb{N}$, premium $P_i \geq 0$ in USD/share, and long per-share payoff $h_i(S_T)$. The per-share totals are given by:

$$\boxed{H_{\mathcal{C}}(S_T) = \sum_{i=1}^{d} \underbrace{\theta_i\, n_i}_{\text{signed copies of leg } i} h_i(S_T), \qquad \Pi_{\mathcal{C}}(S_T) = \sum_{i=1}^{d} \theta_i n_i [h_i(S_T) - P_i]}$$

Both are in USD/share, which is what the example notebooks plot. For account dollars, multiply leg i by its deliverable $q_i \in \mathbb{N}$ shares/contract.

- The kinks are the strikes: each h_i bends only at K_i, so the profit is piecewise linear, with a corner at strike K exactly when the aggregate jump $\sum_{i:K_i=K} \theta_i n_i$ is nonzero. Opposing legs at one strike cancel.
- The offset is the net premium: $\sum_i \theta_i n_i P_i$, the net debit or credit, shifts the diagram without changing its shape.

Legs must share an underlying and an expiration for one diagram in S_T; deliverables may differ, so each q_i is applied to its own leg.

Composite Positions: Three Structures, One Sum

Three structures, D the total premium per share:

- Vertical spread: two legs of one type, different strikes, opposite directions. Both extremes finite.
- Straddle: a call and a put at one strike K, same direction. Profit $\theta(|S_T - K| - D)$, breakevens $K + D$ and, when $D \leq K$, $K - D$.
- Strangle: a call and a put at strikes $K_1 < K_2$, same direction. A flat plateau replaces the straddle's single kink: a trough when long, a peak when short.

State the bounds correctly. A long straddle is routinely said to have "unlimited profit in either direction." That is wrong on the downside. Since $S_T \geq 0$ the put leg cannot pay more than its strike, so the lower branch terminates: the AMD $K = 230$ straddle with $D = 42.88$ caps its downside at 187.12 USD/share, while the upside has no bound.

Example 2: CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

Optional Advanced Material

Advanced: advanced/static_replication/
CHEME-5660-L8b-Advanced-Static-Replication-Fall-2026.ipynb

Optional, not a prerequisite for L9a. The composite rule says adding legs produces payoff shapes; the notebook proves the converse. Every continuous piecewise-linear payoff with kinks $0 < K_1 < \cdots < K_d$ is $a + b\,S_T + \sum_i w_i(S_T - K_i)^+$ in exactly one way, with a the payoff at zero, b the initial slope, and w_i the jump in slope at K_i. It proves existence by a telescoping sum of slopes and uniqueness by sweeping out from the leftmost interval, derives the put payoff identity, tabulates every structure above as one triple (a, b, w), and marks the boundary of what a single expiration can reach.

Notebook: lectures/week-8/L8b/advanced, also linked from the lecture notebook.

Summary

Today we translated contract terms into expiration cash flows and kept the contractual payoff separate from the investor's premium-dependent profit.

Key takeaways

- Payoff belongs to the contract: a long call pays $(S_T - K)^+$ and a long put pays $(K - S_T)^+$ per share, and short positions reverse those flows exactly. Only S_T is uncertain.
- Profit needs a convention: premium, financing time, deliverable q, fees, and direction θ all enter, so strike, moneyness, and breakeven are three different things.
- Loss bounds are asymmetric, and only one side is unlimited: the uncovered short call has no finite bound because S_T has none; a standard short put's worst case is finite at $K - P_0$, reached at $S_T = 0$.

Next, L9a values European claims with Black–Scholes–Merton and American claims with a Cox–Ross–Rubinstein lattice: where the premium comes from.

Today: what the contract promises, what the investor earns, and why only one of the four positions can lose without bound.